YANG LI（郦洋）

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中文简介

上海交通大学人工智能学院在读博士生，专业为计算机科学与技术，导师为鄂维南院士和严骏驰教授，研究方向为广泛的机器学习方法，特别是组合优化与决策、生成式模型、推理大模型等。本科毕业于上海交通大学电子信息与电气工程学院信息安全专业（现计算机学院）。**第一作者/共同一作发表 CCF-A 类顶级会议论文 12 篇，包括机器学习三大顶级会议 NeurIPS、ICML、ICLR 10 篇**。曾获 NeurIPS 和 ICML 的 Spotlight 论文（前 5% 和 3.5%）。主导多个机器学习求解离散优化开源项目、华为重要技术开源、参与阿里巴巴 RL4LLM 框架开发等工作，开源社区共计获超五千星，主导工具包下载超 70k 次。本科、研究生学业排名前 2%，获**国家自然科学基金青年学生基础研究博士生项目（全院唯一）、上海交通大学三好学生标兵称号（全校全年级十人）、中国科协青年科技人才培育工程博士生专项计划（全院唯一）、研究生国家奖学金（系 1%）、本科生国家奖学金（全国 0.2%）**、上海市优秀毕业生（系 5%）、华为奖学金（系 3%）、鹰角奖学金 (系 3%)、上海交大一等优秀奖学金（系 1%）等荣誉。

未来思考方向：技术发展日新月异，深刻的认知沉淀需源自一线基座模型的训练实践。学术研究性质的部分探索易与产业最前沿的实际挑战脱节，未来期望获得更直接的研究验证渠道与认知提升路径，将核心精力投入一线模型训练的关键环节。

主要经历

T-Star 人才计划实习, 阿里巴巴 ATH 事业群 June 2025 – Feb. 2026

Research Intern Mentor: Weixun Wang and Wenbo Su

- **大模型强化学习后训练研究：**1）Attention Illumates LLM Reasoning (ICML 2026 一作) 探索思路“通过更深入地把握模型内部的推理模式，我们能否更有效地强化模型推理能力？”利用 Attention 所揭示的关键推理路径，动态识别推理链中的重要 token 或步骤，并据此进行有针对性的奖励分配。技术报告引起广泛自发报道（超十篇），Hugging Face 单日论文榜单最高第一。2）提出推理调色盘 Reasoning Palette 框架（CVPR 2026 共同一作），旨在通过潜在空间的上下文化实现对大语言模型推理风格的有效控制与训练过程中的可控探索。该方法基于变分自编码器，将问答、思维链、解题方案等输入嵌入到高斯潜在空间中，发现不同推理策略（如数学、代码、问答）在隐空间中自然聚类，从而支持通过采样潜在变量 z 来引导模型生成特定风格或结构的响应。3）提出 FlowTracer (ICML 2026 共同一作) 通过注意力权重构建以 token 为节点的有向无环图，并施加流量守恒约束，提取从问题到答案的信息流骨干，从而实现对 token 级别的精细信用分配。该方法能够精准识别出真正影响推理结果的关键 token，并据此塑造 token 级奖励信号，在多个推理任务上取得了一致的性能提升。
- **RL4LLM 基础设施与智能体模型训练研发：**1）参与贡献阿里巴巴 ALE 智能体学习生态与全链路 ROME 智能体模型开发；ROME 以仅 3B 激活参数 30B 总参在 agentic coding 等真实场景中匹敌 480B+ 模型；ROME+ALE 工作在 X 社区引发讨论（首条推文 134.2K 浏览），被评价为“中国公司挑战 Agent 初创公司幻觉”“从 Prompt Engineering 迈向 Environment Design”的里程碑。2）参与阿里巴巴 RL4LLM 框架 ROLL 的开发，参与贡献异步 RL 训练的高性能系统 ROLL Flash，在 Agentic 任务（如 ALFWorld、SWE）中，最高提速 2.72×，开源社区获 3k 星。

ReThinkLab 实验室, 上海交通大学 Jul. 2021 – Present

Researcher 导师: Prof. Junchi Yan, Prof. Weinan E

- **生成式机器学习范式和生成模型：**提出预测一致性学习 PCL（ICML 2025）将生成式模型的设计思想引入经典预测场景，通过改变标签利用方式重新定义监督学习范式，说明将部分标签信息输入给模型能够帮助模型“参考着答案”学习，在语言生成、图像分割、图问题求解等场景超越传统监督学习。提出 AdvLatGAN（NeurIPS 2022）为生成模型如对抗生成网络等引入一个新的优化维度，即隐变量优化，显著提升生成模型表现，同时从采样角度提出算法解决生成多样性问题（IJCAI 2023），成果获评 NeurIPS22 Spotlight Paper，被评价在扩散模型备受关注的时代重新阐明生成式对抗网络的价值。
- **生成式组合优化求解框架：**提出了性能领先的学习优化求解框架及求解器 T2T (NeurIPS 2023) 和 FastT2T (NeurIPS 2024)，首次提出基于扩散模型的训练到测试两阶段求解框架，同时提出优化场

景下适配的一致性模型，在人工智能组合优化决策的机器学习方法中取得大幅提升，Fast T2T 求解框架在经典问题旅行商问题上对比其他核心先进机器学习求解器达到 82.1% 求解性能优势和 14.7 倍提速。提出 GenSCO（NeurIPS 2025），探索生成式求解框架的 scaling law，重构框架提供针对性技术改进颠覆求解性能瓶颈，在如 TSP-100 问题上，将机器学习求解器求解质量提升十倍、速度提升百倍，相比经典数学求解器达到最优解速度提升百倍。在 MCl 问题上，对比垄断地位的商业求解器 Gurobi，334 倍提速下达到优于其 0.1% 目标函数值的求解结果。
- **机器学习组合优化求解框架整合**：设计机器学习求解路径规划问题、更广泛优化问题的框架，提出 Unify ML4TSP（ICLR 2025）对领域技术进行系统性整合和策略重组，为领域发展提供工具包和指导方向。探索预测、决策联合过程中预测决策一体化方法的框架，重新思考不同场景下预测决策一体化技术的应用（NeurIPS 2024）。主导多个机器学习求解组合优化资源库开源项目，包括 awesome-ml4co,ML4CO-Kit,ML4TSPBench 等，开源社区获超 3k 星，下载超 70k 次。

学习优化求解器项目, with **决策智能实验室，华为诺亚方舟实验室** Apr. 2022 – Feb. 2023

Researcher Leader/Mentor: Xijun Li and Prof. Junchi Yan

- **基于生成式模型的组合优化数据侧支持**：针对基础优化问题如 SAT 问题和 MILP 问题，为缓解工业场景数据瓶颈提出基于图模型的数据生成算法，首次生成计算复杂度近似的优化问题数据，提出的 HardSATGEN (SIGKDD 2023)、MixSATGEN (ICLR 2024) 和 ACM-MILP (ICML 2024) 分别使用基于核心结构和社区结构的图生成算法、图匹配子结构替换算法，以及基于联合约束和特化选择的变分自编码器完成优化数据生成，成果获评 ICML24 Spotlight Paper。算法被集成到华为天筹 OptVerse AI 求解器中，求解器成果在 Hans Mittelmann 国际权威数学优化求解器榜单中排名第一，并在 2023 年世界人工智能大会上获得最高奖 SAIL 奖。
- **求解器贡献**：在传统启发式算法中引入在线学习决策自动化调整算法，独立编码开发了 Kissat-Adaptive-Restart 求解器并在华为海思的 EDA 业务场景中取得了显著成果，平均性能提升约 18%，最大提升达 97%。

Education

Shanghai Jiao Tong University (SJTU), Shanghai, China Mar. 2022 – Present

Doctor (Upgraded from Master's Program), School of Artificial Intelligence
Major: Computer Science and Technology, Supervisor: Weinan E and Junchi Yan

- GPA: 3.83/4.0. Rank Reference: 3 out of 211 achieving the Graduate National Scholarship
- Courses: 90.0% at A level

Shanghai Jiao Tong University (SJTU), Shanghai, China Sep. 2018 – Jul. 2022

Bachelor, Cyber Science and Engineering, School of Electronic Information and Electrical Engineering (Now School of Computer Science)

- GPA: 91.03/100 (or 3.93/4.3), Rank: 3/129
- Foundation Courses: 73.33% above A, 40.00% above A+
- Subject Courses: 80% above A, 50.00% above A+

Publications

First-authored (in Chronological Order):

1. **How Does Reasoning Flow? Tracing Attention-Induced Information Flow for Targeted RL in LLMs**
Zhichen Dong*, Yang Li*, Yuhan Sun, Weixun Wang, Yijia Luo, Zinian Peng, Wenbo Su, YuCheng, Bo Zheng, Junchi Yan (*: equal contribution)
International Conference on Machine Learning, ICML 2026
2. **Attention Illuminates LLM Reasoning: The Preplan-and-Anchor Rhythm Enables Fine-Grained Policy Optimization**
Yang Li, Zhichen Dong, Yuhan Sun, Weixun Wang, Shaopan Xiong, Yijia Luo, Jiashun Liu, Han Lu, Jiamang Wang, Wenbo Su, Bo Zheng, Junchi Yan
International Conference on Machine Learning, ICML 2026
3. **Reasoning Palette: Modulating Reasoning via Latent Contextualization for Controllable Exploration for (V)LMs**

Rujiao Long*, Yang Li*, Xingyao Zhang, Weixun Wang, Tianqianjin Lin, Xi Zhao, Yuchi Xu, Wenbo Su, Junchi Yan, Bo Zheng
Conference on Computer Vision and Pattern Recognition, CVPR 2026

4. **MaskCO: Masked Generation Drives Effective Representation Learning and Exploiting for Combinatorial Optimization**
   Lvda Chen*, Yang Li*, Junchi Yan
   International Conference on Learning Representations, ICLR 2026
5. **Generation as Search Operator for Test-Time Scaling of Diffusion-based Combinatorial Optimization**
   Yang Li, Lvda Chen, Haonan Wang, Runzhong Wang, Junchi Yan
   Advances in Neural Information Processing Systems, NeurIPS 2025
6. **Generative Modeling Reinvents Supervised Learning: Label Repurposing by Predictive Consistency Learning**
   Yang Li, Jiale Ma, Yebin Yang, Qitian Wu, Hongyuan Zha, Junchi Yan
   Forty-Second International Conference on Machine Learning ICML 2025
7. **Unify ML4TSP: Drawing Methodological Principles for TSP and Beyond from Streamlined Design Space of Learning and Search**
   Yang Li, Jiale Ma, Wenzheng Pan, Runzhong Wang, Haoyu Geng, Nianzu Yang, Junchi Yan
   International Conference on Learning Representations, ICLR 2025
8. **Fast T2T: Optimization Consistency Speeds Up Diffusion-Based Training-to-Testing Solving for Combinatorial Optimization**
   Yang Li, Jinpei Guo, Runzhong Wang, Hongyuan Zha, Junchi Yan
   Advances in Neural Information Processing Systems, NeurIPS 2024
9. **MixSATGEN: Learning Graph Mixing for SAT Instance Generation**
   Xinyan Chen*, Yang Li*, Runzhong Wang, Junchi Yan (*: equal contribution)
   International Conference on Learning Representations, ICLR 2024
10. **T2T: From Distribution Learning in Training to Gradient Search in Testing for Combinatorial Optimization**
    Yang Li, Jinpei Guo, Runzhong Wang, Junchi Yan
    Advances in Neural Information Processing Systems, NeurIPS 2023
11. **HardSATGEN: Understanding the Difficulty of Hard SAT Formula Generation and A Strong Structure-Hardness-Aware Baseline**
    Yang Li, Xinyan Chen, Wenxuan Guo, Xijun Li, Wanqian Luo, Junhua Huang, Hui-Ling Zhen, Mingxuan Yuan, Junchi Yan
    SIGKDD Conference on Knowledge Discovery and Data Mining, KDD 2023
12. **IID-GAN: an IID Sampling Perspective for Regularizing Mode Collapse**
    Yang Li*, Liangliang Shi*, Junchi Yan (*: equal contribution)
    International Joint Conference on Artificial Intelligence, IJCAI 2023
13. **Improving Generative Adversarial Networks via Adversarial Learning in Latent Space**
    Yang Li, Yichuan Mo, Liangliang Shi, Junchi Yan
    Advances in Neural Information Processing Systems, NeurIPS 2022 (Spotlight Top 5%)
14. **Kissat_Adaptive_Restart, Kissat_Cfexp: Adaptive Restart Policy and Variable Scoring Improvement**
    Yang Li, Yuqi Jia, Wanqian Luo, Hui-Ling Zhen, Xijun Li, Mingxuan Yuan, Junchi Yan
    Proceedings of SAT Competition 2022

Other (in Chronological Order):

1. **Problem Distributions as Tasks: Repurposing Meta Learning for Generative Combinatorial Optimization towards Multi-task Pretrain and Adaptation**
   Wenzheng Pan, Jiale Ma, Nuoyan Chen, Yang Li, Junchi Yan
   International Conference on Machine Learning, ICML 2026
2. **ConRep4CO: Contrastive Representation Learning of Combinatorial Optimization Instances across Types**
   Ziao Guo, Yang Li, Shiyue Wang, Junchi Yan
   International Conference on Learning Representations, ICLR 2026
3. **Native Adaptive Solution Expansion for Diffusion-based Combinatorial Optimization**

Yu Wang, Yang Li, Jiale Ma, Junchi Yan, Yi Chang
International Conference on Learning Representations, ICLR 2026

4. **Smarter Not Harder: Generative Process Evaluation with Intrinsic-Signal Driving and Ability Adaptive Reward Shaping**
   Tao He, Rongchuan Mu, Lizi Liao, Yixin Cao, Yang Li, Yijia Luo, Weixun Wang, Ming Liu, Bing Qin
   International Conference on Learning Representations, ICLR 2026
5. **Let It Flow: Agentic Crafting on Rock and Roll, Building the ROME Model within an Open Agentic Learning Ecosystem**
   ROCK & ROLL & IFLOW & DT Joint Team (Core Contributor)
   Technical Report (arXiv), 2026
6. **Part II: ROLL Flash–Accelerating RLVR and Agentic Training with Asynchrony**
   Han Lu, Zichen Liu, Shaopan Xiong, Yancheng He, Wei Gao, Yanan Wu, Weixun Wang, Jiashun Liu, Yang Li, Haizhou Zhao, Ju Huang, Siran Yang, Xiaoyang Li, Yijia Luo, Zihe Liu, Ling Pan, Junchi Yan, Wei Wang, Wenbo Su, Jiamang Wang, Lin Qu, Bo Zheng
   Technical Report (arXiv), 2025
7. **StruDiCO: Structured Denoising Diffusion with Gradient-free Inference-stage Boosting for Memory and Time Efficient Combinatorial Optimization**
   Yu Wang, Yang Li, Junchi Yan, Yi Chang
   Advances in Neural Information Processing Systems, NeurIPS 2025
8. **ML4CO-Bench-101: Benchmark Machine Learning for Classic Combinatorial Problems on Graphs**
   Jiale Ma, Wenzheng Pan, Yang Li, Junchi Yan
   Advances in Neural Information Processing Systems, NeurIPS 2025
9. **Bridging Crypto with ML-based Solvers: the SAT Formulation and Benchmarks**
   Xinhao Zheng, Xinhao Song, Bolin Qiu, Yang Li, Zhongteng Gui, Junchi Yan
   Advances in Neural Information Processing Systems, NeurIPS 2025
10. **COExpander: Adaptive Solution Expansion for Combinatorial Optimization**
    Jiale Ma, Wenzheng Pan, Yang Li, Junchi Yan
    Forty-Second International Conference on Machine Learning ICML 2025
11. **UniCO: On Unified Combinatorial Optimization via Problem Reduction to Matrix-Encoded General TSP**
    Hao Xiong, Wenzheng Pan, Jiale Ma, Wentao Zhao, Yang Li, Junchi Yan
    International Conference on Learning Representations, ICLR 2025
12. **Learning Plaintext-Ciphertext Cryptographic Problems via ANF-based SAT Instance Representation**
    Xinhao Zheng, Yang Li, Cunxin Fan, Huaijin Wu, Xinhao Song, Junchi Yan
    Advances in Neural Information Processing Systems, NeurIPS 2024
13. **There is No Silver Bullet: Benchmarking Methods in Predictive Combinatorial Optimization**
    Haoyu Geng, Han Ruan, Runzhong Wang, Yang Li, Yang Wang, Lei Chen, Junchi Yan
    Advances in Neural Information Processing Systems, NeurIPS 2024
14. **正线性约束组合优化问题的非自回归学习求解 | Learning to Solve Combinatorial Optimization under Positive Linear Constraints via Non-Autoregressive Neural Networks**
    汪润中, 郦洋, 严骏驰, 杨小康 | Runzhong Wang, Yang Li, Junchi Yan, Xiaokang Yang
    中国科学：信息科学 | SCIENTIA SINICA Informationis 2024.
15. **ACM-MILP: Adaptive Constraint Modification via Grouping and Selection for Hardness-Preserving MILP Instance Generation**
    Ziao Guo, Yang Li, Chang Liu, Wenli Ouyang, Junchi Yan
    International Conference on Machine Learning, ICML 2024 (Spotlight Top 3.5%)
16. **Molecule Generation for Drug Design: a Graph Learning Perspective**
    Nianzu Yang, Huaijin Wu, Kaipeng Zeng, Yang Li, Junchi Yan
    Fundamental Research, 2024
17. **Machine Learning Insides OptVerse AI Solver: Design Principles and Applications**
    Xijun Li, Fangzhou Zhu, Hui-Ling Zhen, Weilin Luo, Meng Lu, Yimin Huang, Zhenan Fan, Zirui Zhou, Yufei Kuang, Zhihai Wang, Zijie Geng, Yang Li, Haoyang Liu, Zhiwu An, Muming Yang, Jianshu Li, Jie Wang, Junchi Yan, Defeng Sun, Tao Zhong, Yong Zhang, Jia Zeng, Mingxuan Yuan, Jianye Hao, Jun Yao, Kun Mao
    arXiv preprint, 2024

18. **The Policy-gradient Placement and Generative Routing Neural Networks for Chip Design**
Ruoyu Cheng, Xianglong Lyu, Yang Li, Junjie Ye, Jianye Hao, Junchi Yan
Advances in Neural Information Processing Systems, NeurIPS 2022

Honors and Awards

National Natural Science Foundation of China (NSFC) Youth Student Fundamental Research Program −Doctoral Fellowship (the only recipient in the School)	2025
China Association for Science and Technology (CAST) Young Talent Development Program −Doctoral Fellowship (the only recipient in the School)	2025
Shanghai Jiao Tong University Pacemaker to Merit Student Award (Top 10 university-wide)	2025
National Scholarship (top 0.2% in the nation)	2019
Graduate National Scholarship (top 1% in the CS department)	2023
Outstanding Graduate of Shanghai (top 3%)	2022
Huawei Fellowship (top 3%)	2020
HyperGryph Fellowship (top 3%)	2021
1st-Class Academic Excellence Scholarship (top 1%)	2019
NeurIPS 2024 Top Reviewer	2024
Merit Student of Shanghai Jiao Tong University	2019, 2020
1st-Class Academic Scholarship for Graduate Students	2022
Special Prize for Social Practice of SJTU	2020
First Prize for Social Practice of SJTU	2019
Advanced Individuals in Social Practice of SJTU	2020

Miscellaneous

- Academic Service: I serve as the reviewer for top-tier ML conferences, e.g. NeurIPS, ICML, ICLR, and journals, e.g. TPAMI.
- Blog: https://yangco-le.github.io/
- GitHub: https://github.com/yangco-le
- Languages: English - Experienced, Mandarin - Native speaker